JAVA PROJECT
(Intercity Bus Reservation System)

Abstract

The Intercity Bus Reservation System is a Java-based console application designed to simulate the process of searching, selecting, booking, and managing bus reservations between different cities. The main objective of this project is to develop a simple and practical reservation system while strengthening the programming fundamentals and logical thinking skills of first-year BTech students. The system allows users to search for available buses, choose a route and travel date, view the available seat layout, book seats, cancel reservations, and view their reservation details. The project is designed using basic Java concepts so that it can be easily understood, modified, and expanded as students learn more programming concepts. The system begins by allowing the user to select a source city, destination city, and travel date. Based on the selected route, the program can display suitable buses and their basic details, such as bus number, departure time, arrival time, and ticket price. After selecting a bus, the user can view a seat map showing which seats are available and which have already been booked. The user can then select one or more available seats for reservation. Before confirming a booking, the system checks whether the entered seat number is valid and whether the selected seat is already booked. If the seat is available, the reservation is confirmed and the seat status is updated. If the seat is already booked, the system displays an appropriate message and allows the user to choose another seat. The system also provides a cancellation feature through which users can cancel a previously booked seat. During cancellation, the program verifies whether the selected seat is actually reserved before changing its status back to available. The system can also calculate the total ticket cost based on the number of seats selected and the ticket price of the chosen bus. The project primarily focuses on the use of control statements and is structured so that additional programming concepts can be added later without completely changing the existing program. The if-else statement can be used for important decision-making tasks, such as checking seat availability, validating route selection, and handling incorrect inputs. A switch statement can be used to create a clear booking menu through which users can select options such as searching for a bus, viewing seats, booking a seat, cancelling a reservation, and viewing booking details. A for loop can be used to display the bus seat map and process multiple seats, while a while loop can keep the reservation process running until the user chooses to exit. Arrays can be used to store seat availability and other basic information. The program can also use variables, operators, Scanner input, and simple calculations to handle user choices and reservation

data. One of the major advantages of this project is that it can be developed gradually. In the initial version, students can implement basic if-else conditions, loops, arrays, and user input. Later, they can add methods to organize the program, classes and objects to introduce object-oriented programming, multiple buses and routes, different seat categories, passenger details, fare calculation, booking IDs, and improved input validation. Additional features such as displaying available buses based on the selected route and date, generating a simple ticket or receipt, maintaining booking history, and calculating discounts can also be incorporated as students gain more knowledge. Overall, the Intercity Bus Reservation System provides a realistic and manageable problem for a first-year BTech programming project. It combines real-world reservation requirements with fundamental programming concepts and provides sufficient scope for logical problem-solving, experimentation, and future improvement. The project is neither overly complex nor too basic, making it suitable for demonstrating a student's understanding of control statements while providing a clear path toward more advanced Java programming concepts. Through this project, students can gain practical experience in designing program flow, handling user input, making decisions, processing repeated operations, managing data, and developing a functional application step by step. The final system can therefore serve as a strong foundation for future projects involving databases, graphical interfaces, and larger reservation management systems.

Keywords:

- **if-else → seat availability, validation**
- **switch → booking menu**
- **for loop → seat map**
- **while loop → continue reservation process**
- **Scanner → user input**
- **1D-Arrays → seat storage**

CODE:

```
import java.util.Scanner;

public class BusReservationSystemss {

    static Scanner sc = new Scanner(System.in);
    static double[][] seats = new double[4][5];

    static String busNumber = "";
    static String route = "";
    static String travelDate = "";
    static String departureTime = "";
    static String arrivalTime = "";
    static double ticketPrice = 0;
    static double totalAmount = 0;
    static int bookedSeats = 0;

    // Method to display seat map
    static void displaySeats() {
        System.out.println("\n========== SEAT MAP ==========");

        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 5; j++) {

                int seatNo = i * 5 + j + 1;

                if (seats[i][j] == 0)
                    System.out.print(seatNo + "[A]  ");
                else
                    System.out.print(seatNo + "[X]  ");
            }
            System.out.println();
        }

        System.out.println("A = Available   X = Booked");
    }

    // Method to book a seat
    static void bookSeat() {
        if (route.equals("")) {
            System.out.println("Please search a bus first.");
            return;
        }

        System.out.print("Enter seat number (1-20): ");
        int seatNo = sc.nextInt();
```

```
    if (seatNo < 1 || seatNo > 20) {
        System.out.println("Invalid seat number.");
        return;
    }

    int row = (seatNo - 1) / 5;
    int col = (seatNo - 1) % 5;

    if (seats[row][col] != 0) {
        System.out.println("Seat already booked.");
        return;
    }

    System.out.println("Ticket Price: Rs. " + ticketPrice);
    System.out.print("Confirm booking? (1-Yes / 0-No): ");
    int confirm = sc.nextInt();

    if (confirm == 1) {
        seats[row][col] = ticketPrice;
        bookedSeats++;
        totalAmount += ticketPrice;

        System.out.println("Seat " + seatNo + " booked successfully!");
    } else {
        System.out.println("Booking cancelled.");
    }
}

// Method to cancel a seat
static void cancelSeat() {
    System.out.print("Enter seat number to cancel: ");
    int seatNo = sc.nextInt();

    if (seatNo < 1 || seatNo > 20) {
        System.out.println("Invalid seat number.");
        return;
    }

    int row = (seatNo - 1) / 5;
    int col = (seatNo - 1) % 5;

    if (seats[row][col] == 0) {
        System.out.println("Seat is not booked.");
        return;
    }

    totalAmount -= seats[row][col];
    seats[row][col] = 0;
```

```
    bookedSeats--;

    System.out.println("Reservation cancelled successfully.");
}

public static void main(String[] args) {

    int choice = 0;

    while (choice != 6) {

        System.out.println("\n================================");
        System.out.println(" INTERCITY BUS RESERVATION SYSTEM");
        System.out.println("================================");
        System.out.println("1. Search Bus");
        System.out.println("2. Display Seat Map");
        System.out.println("3. Book Seat");
        System.out.println("4. Cancel Reservation");
        System.out.println("5. View Reservation");
        System.out.println("6. Exit");

        System.out.print("Enter choice: ");
        choice = sc.nextInt();

        switch (choice) {

            case 1:
                System.out.println("\n1. Hyderabad to Bangalore");
                System.out.println("2. Hyderabad to Chennai");
                System.out.println("3. Hyderabad to Vijayawada");

                System.out.print("Select route: ");
                int r = sc.nextInt();

                if (r == 1) {
                    route = "Hyderabad to Bangalore";
                    busNumber = "TS1010";
                    departureTime = "08:00 PM";
                    arrivalTime = "07:50 AM";
                    ticketPrice = 950;
                }
                else if (r == 2) {
                    route = "Hyderabad to Chennai";
                    busNumber = "TS2026";
                    departureTime = "09:00 PM";
                    arrivalTime = "08:15 AM";
                    ticketPrice = 1499;
                }
```

```
            else if (r == 3) {
                route = "Hyderabad to Vijayawada";
                busNumber = "TS6048";
                departureTime = "07:00 PM";
                arrivalTime = "09:00 AM";
                ticketPrice = 899;
            }
            else {
                System.out.println("Invalid route.");
                break;
            }

            System.out.print("Enter travel date: ");
            travelDate = sc.next();

            System.out.println("\nBus Found!");
            System.out.println("Bus Number : " + busNumber);
            System.out.println("Route      : " + route);
            System.out.println("Date       : " + travelDate);
            System.out.println("Departure  : " + departureTime);
            System.out.println("Arrival    : " + arrivalTime);
            System.out.println("Price      : Rs. " + ticketPrice);
            break;

        case 2:
            displaySeats();
            break;

        case 3:
            bookSeat();
            break;

        case 4:
            cancelSeat();
            break;

        case 5:
            System.out.println("\n========== RESERVATION
==========");

            if (bookedSeats == 0) {
                System.out.println("No seats booked.");
            } else {
                System.out.println("Bus Number : " + busNumber);
                System.out.println("Route      : " + route);
                System.out.println("Travel Date: " + travelDate);
                System.out.println("Booked Seats: " + bookedSeats);
                System.out.println("Total Amount: Rs. " + totalAmount);
```

```
                }
                break;

            case 6:
                System.out.println("Thank you for using the system!");
                break;

            default:
                System.out.println("Invalid choice.");
        }
    }

    sc.close();
  }
}
```

Output We Achived:

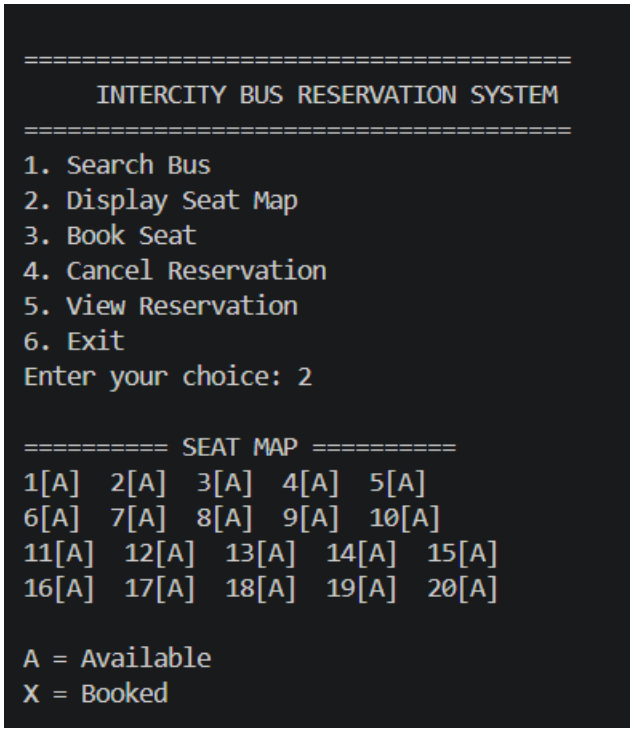
To Search the Bus:

```
=====================================
     INTERCITY BUS RESERVATION SYSTEM
=====================================
1. Search Bus
2. Display Seat Map
3. Book Seat
4. Cancel Reservation
5. View Reservation
6. Exit
Enter your choice: 1

========== SEARCH BUS ==========
1. Hyderabad to Bangalore
2. Hyderabad to Chennai
3. Hyderabad to Vijayawada
Select route: 1
Enter travel date: 2/4/20206

Bus Found!
------------------------------
Bus Number   : TS1010
Route        : Hyderabad to Bangalore
Travel Date  : 2/4/20206
Departure    : 08:00 PM
Arrival      : 07:50 AM
Ticket Price : Rs. 950.0
------------------------------
```

To Display Seat:



Book Seat:

```
=======================================
     INTERCITY BUS RESERVATION SYSTEM
=======================================
1. Search Bus
2. Display Seat Map
3. Book Seat
4. Cancel Reservation
5. View Reservation
6. Exit
Enter your choice: 3

Enter seat number (1-20): 2

Seat 2 is available.
Ticket Price: Rs. 950.0
Confirm booking? (1 = Yes, 0 = No): 1

Seat booked successfully!
Bus Number : TS1010
Route      : Hyderabad to Bangalore
Date       : 2/4/20206
Seat       : 2
Amount     : Rs. 950.0
```

For cancellation:

```
=====================================
     INTERCITY BUS RESERVATION SYSTEM
=====================================
1. Search Bus
2. Display Seat Map
3. Book Seat
4. Cancel Reservation
5. View Reservation
6. Exit
Enter your choice: 4

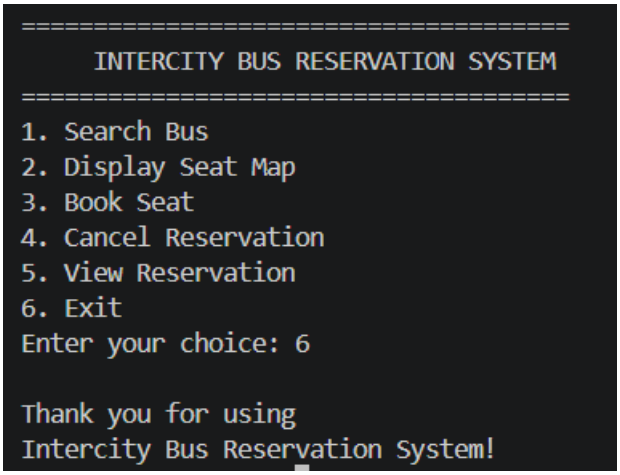
Enter seat number to cancel: 2
Reservation cancelled successfully.
```

For Viewing Reservation:

```
=====================================
     INTERCITY BUS RESERVATION SYSTEM
=====================================
1. Search Bus
2. Display Seat Map
3. Book Seat
4. Cancel Reservation
5. View Reservation
6. Exit
Enter your choice: 5

========== RESERVATION DETAILS ==========
No seats are currently booked.
```

6.Exit


INTERCITY BUS RESERVATION SYSTEM
1. Search Bus
2. Display Seat Map
3. Book Seat
4. Cancel Reservation
5. View Reservation
6. Exit
Enter your choice: 6
Thank you for using
Intercity Bus Reservation System!